Amazon Web Services: Amazon Virtual Private Cloud (VPC)

Topics

- VPC Introduction
- Fully Customizable Network
- VPNs
- Network Diagram
- VPC Features
- Comparing VPCs
- Demo
- Using NAT Gateways for Internet Access
- Demo
- Protecting Your Resources with Security Groups

Topics

- Controlling Subnet Traffic with Network ACLs
- Demo
- VPC Endpoints
- Demo
- VPC Peering
- Demo
- AWS PrivateLink
- Direct Connect

What is a VPC?

- Virtual data center in the cloud
- Logically isolated part of AWS Cloud where you define your own network
- Complete control of the virtual network, your own IP address range, subnets, route tables, and network gateways

Amazon Virtual Private Cloud (Amazon VPC)

Fully Customizable Network

Web

Public-facing subnet

Public subnet

Application

Private subnet. Can only speak to the web tier and database tier

Private subnet

Database

Private subnet. Can only speak to the application tier

Private subnet

What is a VPN?

- Virtual private network connection between on-prem data center and VPC
- Virtual = no physical cables involved in the connection process
- Private = through this connection no one else can see your data or browsing activity
- Networked = multiple devices – your computer and the VPN server – work together to maintain an established connection

VPN connection

Demo: CIDR.xyz

CIDR.xyz

AN INTERACTIVE IP ADDRESS AND CIDR RANGE VISUALIZER

CIDR is a notation for describing blocks of IP addresses and is used heavily in various networking configurations. IP addresses contain 4 octets, each consisting of 8 bits giving values between 0 and 255. The decimal value that comes after the slash is the number of bits consisting of the routing prefix. This in turn can be translated into a netmask, and also designates how many available addresses are in the block.

10 . 88 . 135 . 144 / 28

0 0 0 0 1 0 1 0	0 1 0 1 1 0 0 0	1 0 0 0 0 1 1 1	1 0 0 1 0 0 0 0

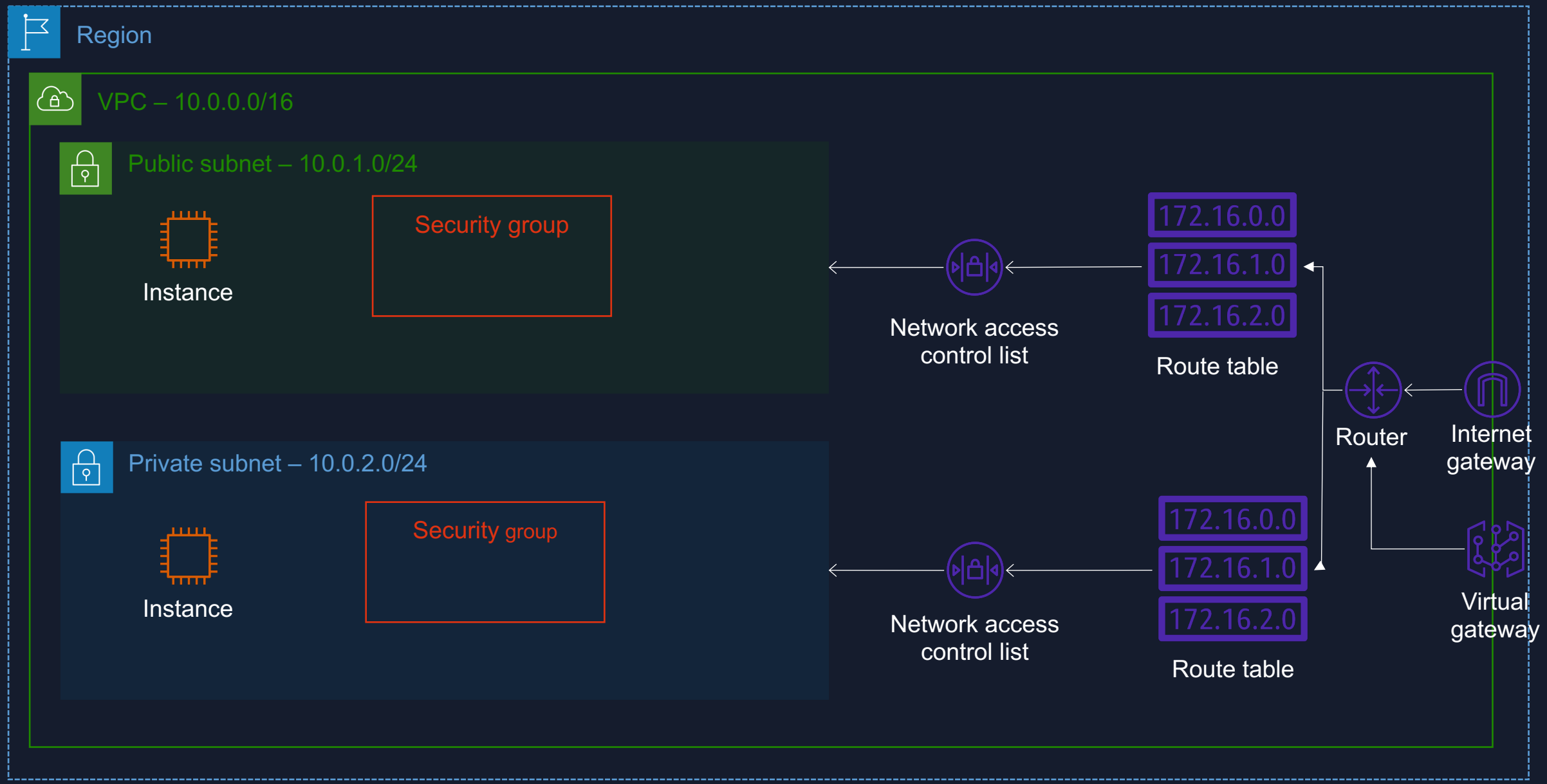
255.255.255.240	10.88.135.144	10.88.135.159	16
NETMASK	CIDR BASE IP	BROADCAST IP	COUNT

10.88.135.145	10.88.135.158
FIRST USABLE IP	LAST USABLE IP

* For routing mask values <= 30, first and last IPs are base and broadcast addresses and are unusable.

Created by Yuval Adam. Source available on Github.

Network Diagram



VPC Features

Launch instances

Custom IP Addresses

172.16.0.0
172.16.1.0
172.16.2.0

Route Tables

Internet Gateway

More Control

Access Control Lists

Use Network Access Control Lists to block specific IP addresses

Comparing VPCs

Default VPC vs. Custom VPC

Default

- User Friendly
- All subnets in default VPC have a route out to the internet
- Each EC2 instance has a public and private IP address

VS

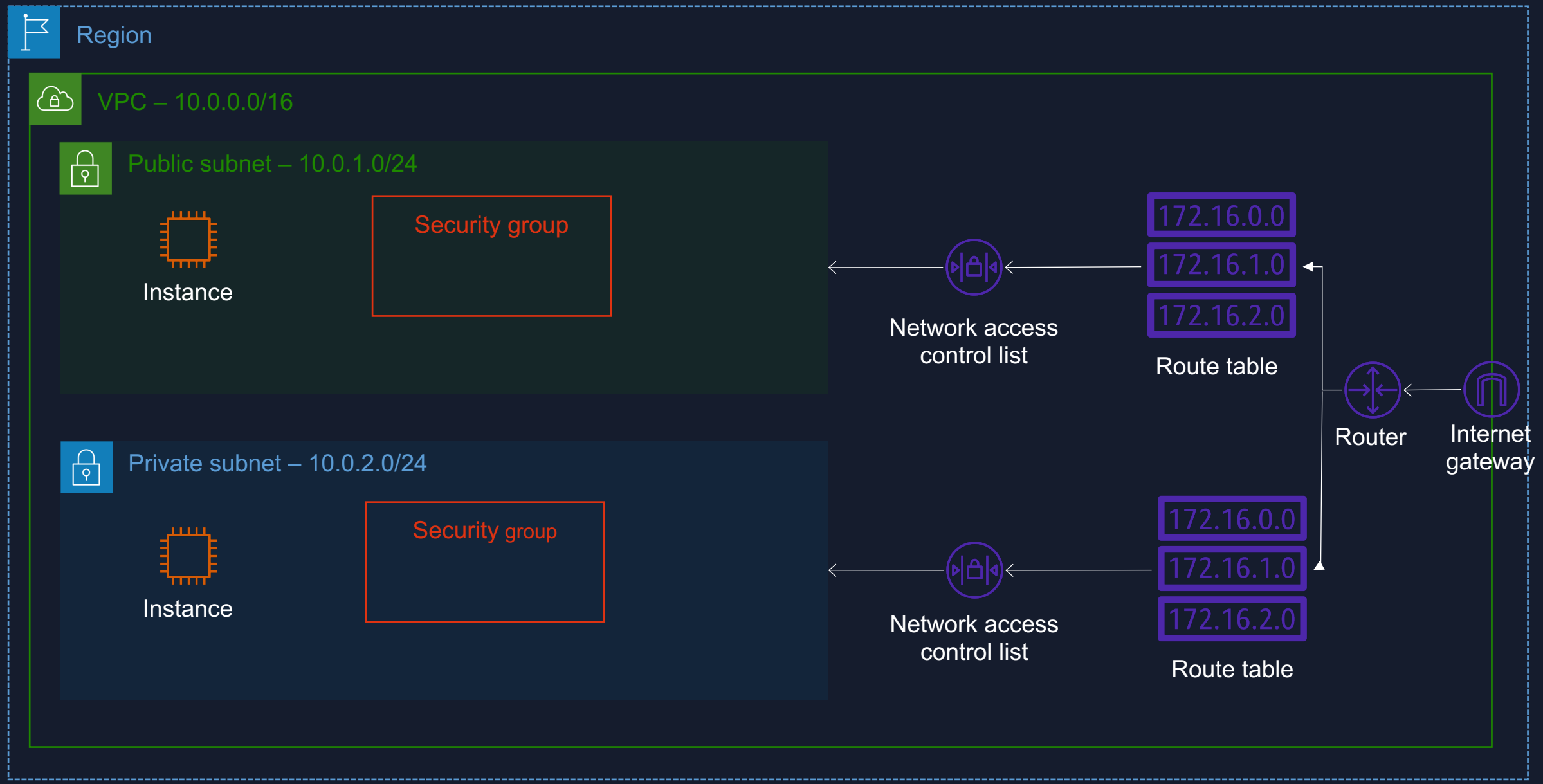
Custom

- Fully Customizable
- Takes time to set up

Demo: Provisioning a VPC

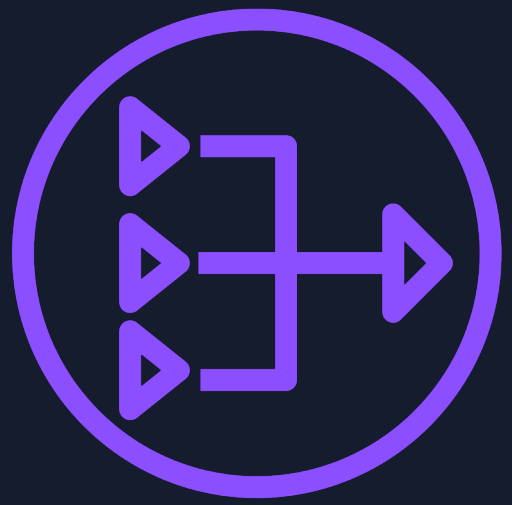
Network Diagram

Network Diagram



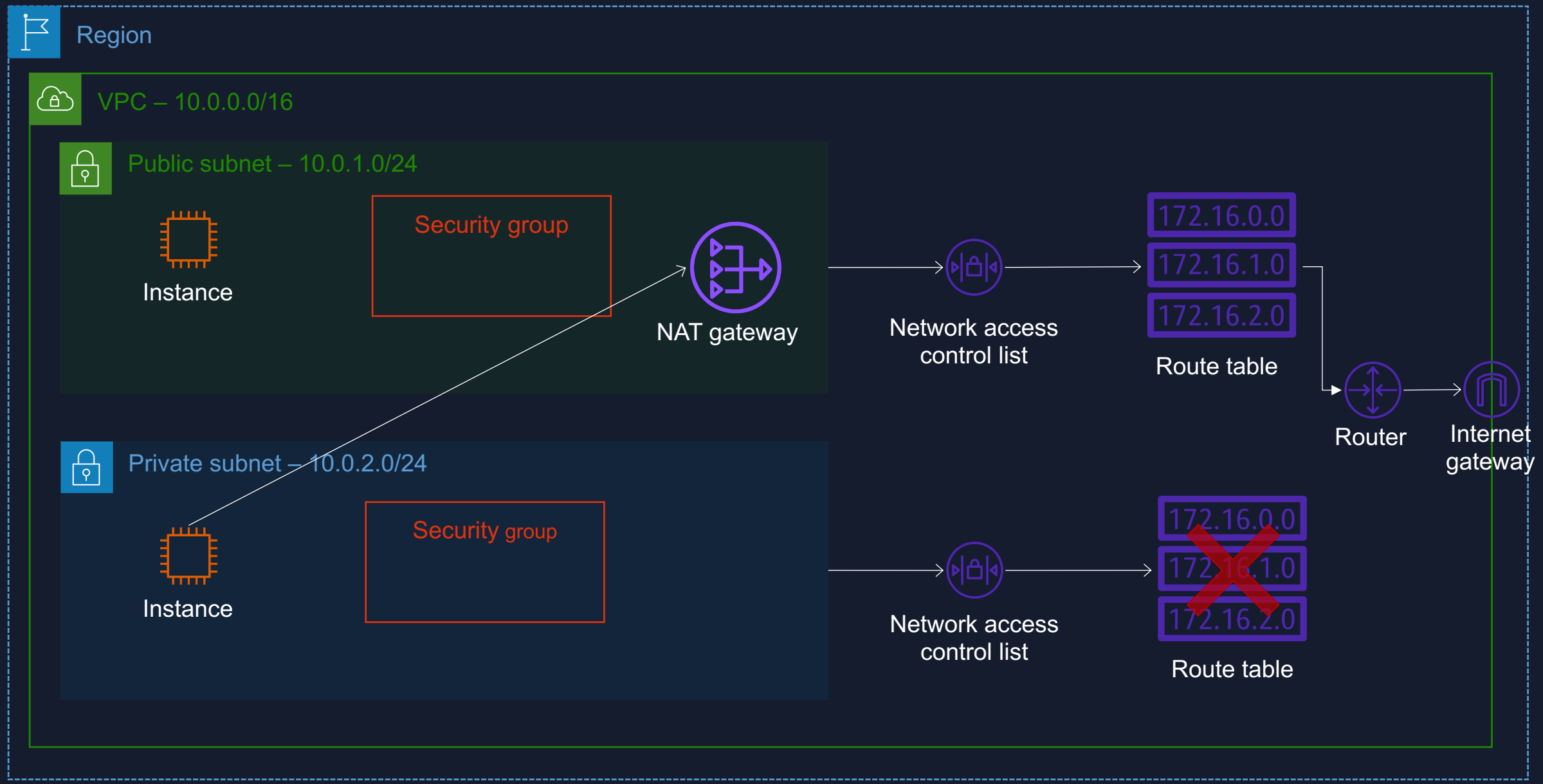
Using NAT Gateways for Internet Access

- Stands for Network Address Translation (NAT) gateway
- Enable instances in a private subnet to connect to the internet or other AWS services
- Prevents the internet from initiating a connection with those instances



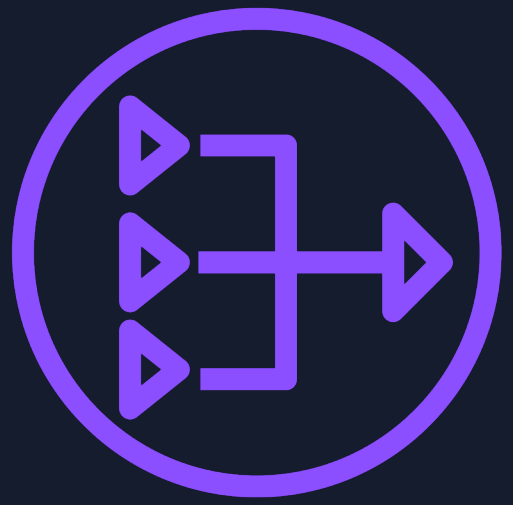
NAT gateway

Network Diagram



Demo: NAT Gateway Facts

- Redundant inside the AZ
- No need to patch
- Not associated with security groups
- Automatically assigned a public IP address



NAT gateway

Protecting Your Resources with Security Groups

How Computers Communicate

Linux
SSH
Port 22

Windows
RDP
Port 3389

HTTP
WEB
BROWSING
Port 80

HTTPS
ENCRYPTED
WEB
BROWSING
(SSL)
Port 443

Security Groups

- Virtual firewalls for you EC2 instances - last line of defense that traffic hits
- Can place any number of EC2 instances with a security group
- Can have multiple security groups attached to an EC2 instance
- By default, all inbound traffic is blocked, and all outbound traffic is allowed
- To let everything in use IP address range 0.0.0.0/0
- In production, you'd only open 0.0.0.0/0 to port 80 (HTTP) and to port 443 (HTTPS)
- Stateful
  - if you send a request from your EC2 instance, the response traffic for that request is allowed to flow in regardless of inbound security groups rules
  - Responses to allowed inbound traffic are allowed to flow out, regardless of outbound rules

Controlling Subnet Traffic with Network ACLs

Network ACLs

- The first line of defense
- Optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets
- Might set up network ACLs with rules like your security groups to add another layer of security to your VPC
- VPC automatically comes with a default network ACL
  - Allows all outbound and inbound traffic
- Can create custom network ACLs
  - By default, denies all inbound and outbound traffic until you add rules
- Each subnet in your VPC must be associated with a network ACL
  - If not, subnet is automatically associated with the default network ACL
- Block IP addresses using network ACL and not security groups

Network ACLs

- You can associate a network ACL with multiple subnets
- A subnet can be associated with only 1 network ACL at a time
- Consists of a numbered list of rules that are evaluated in order starting with the lowest numbered rule
- Have separate inbound and outbound rules and each rule can either allow or deny traffic
- Are stateless – responses to allowed inbound traffic are subject to the rules of outbound traffic (and vice versa)

Demo: Network ACLs

VPC Endpoints

VPC Endpoints

- Enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by AWS PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect
- Amazon VPC instances do not require public IP addresses to communicate with resources of the service
- Traffic between an Amazon VPC and a service does not leave the Amazon network
- Endpoints are virtual devices

https://docs.aws.amazon.com/whitepapers/latest/aws-privatelink/what-are-vpc-endpoints.html

2 Types of Endpoints

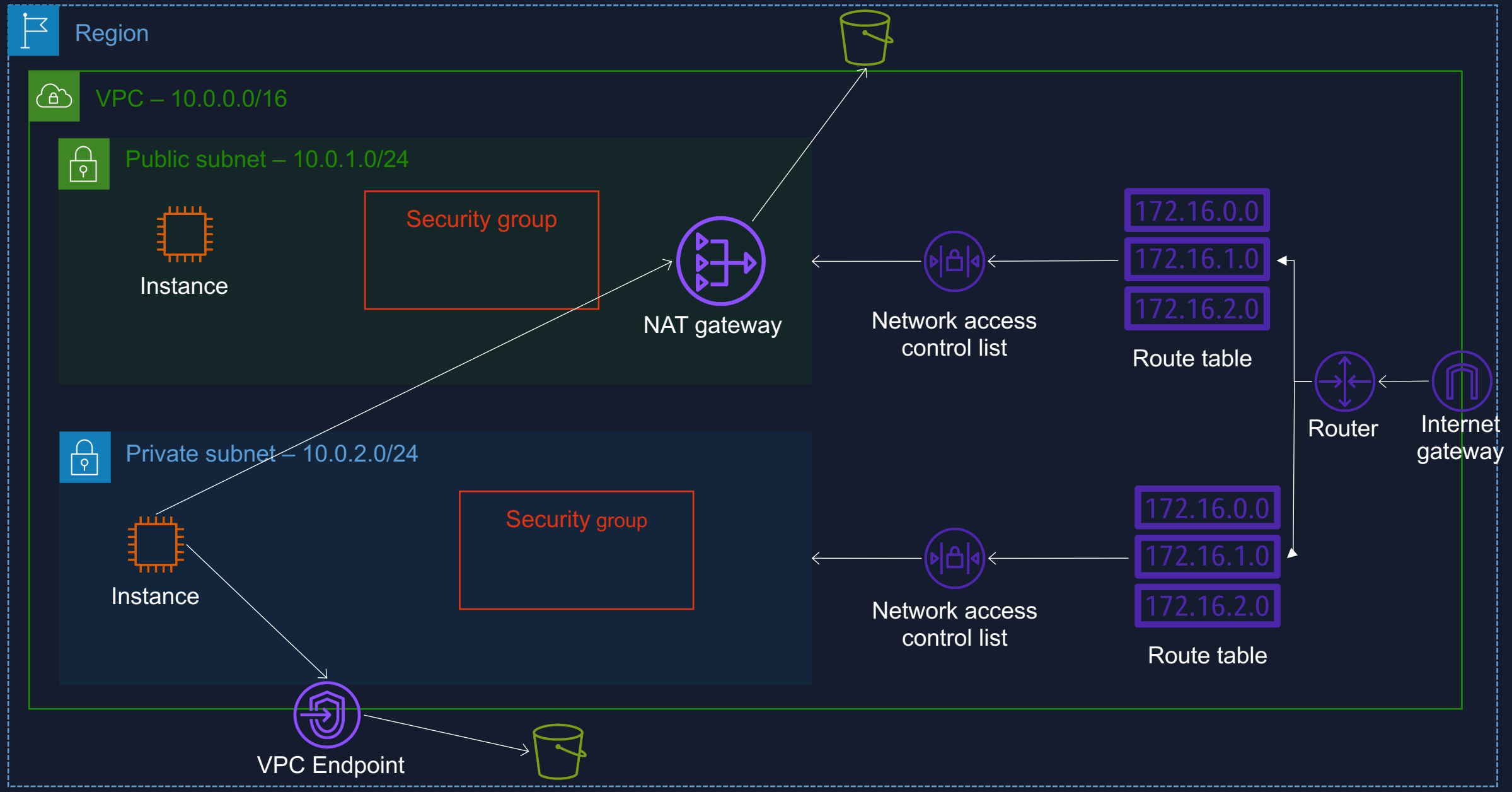
OPTION 1 Interface Endpoints

- Elastic network interface with a private IP address that serves as an entry point for traffic headed to a supported service over AWS PrivateLink
- Supports many AWS services

OPTION 2 Gateway Endpoints

- Similar to NAT Gateways
- Target specific IP routes in a route table
- Do not enable PrivateLink
- Provisioned virtual device
- Supports connection to S3 and DynamoDB

Network Diagram

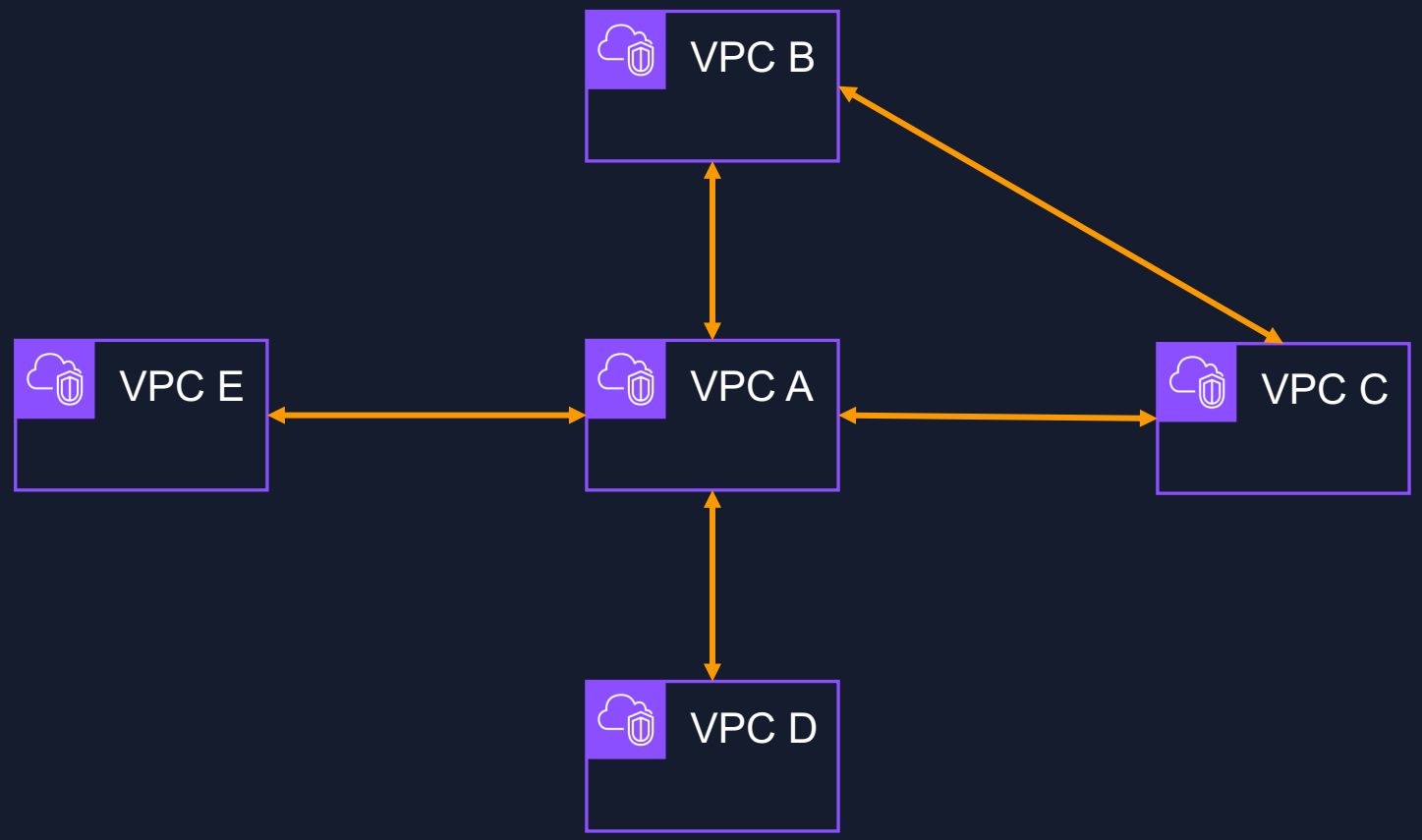


VPC Peering

VPC Peering

- Allows you to connect 1 VPC with another via a direct network route using private IP addresses
- Instances behave as if they were on the same private network
- Can peer VPCs with other AWS accounts and with other VPCs in the same account
- Peering is in a star configuration (e.g., 1 central VPC with 4 other VPCs) i.e., no transitive peering
- You can peer between regions

Transitive Peering

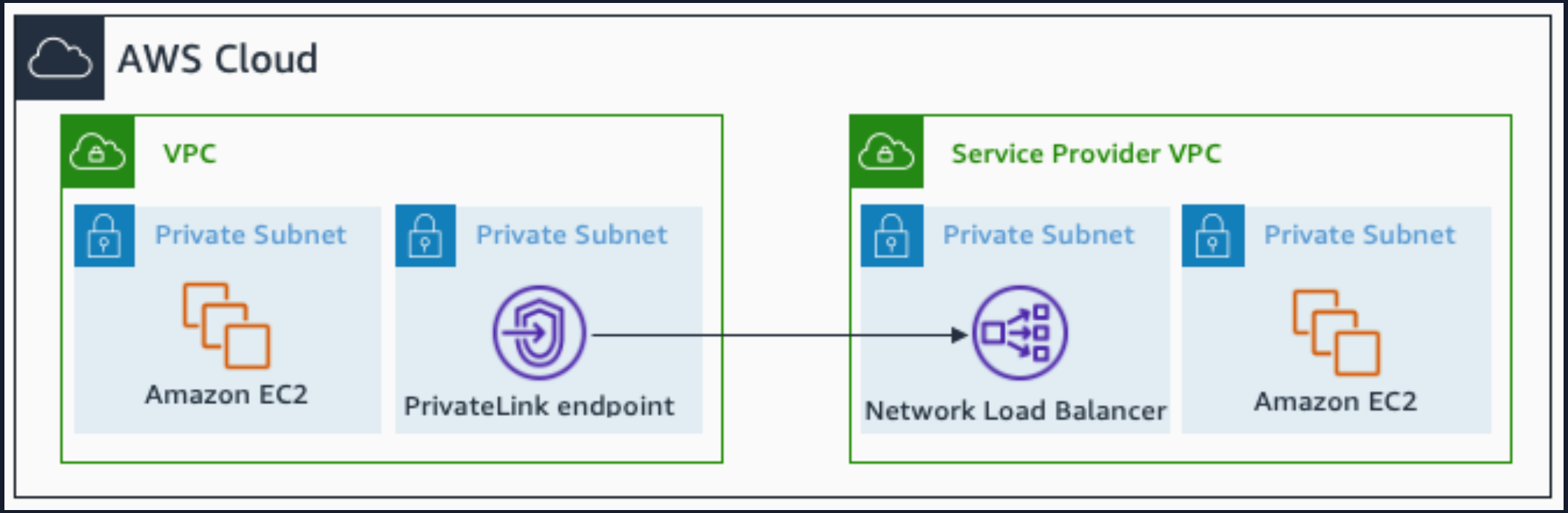


AWS PrivateLink

Opening Services in a VPC to Another VPC

1. Open up VPC to internet
   - Security Considerations – everything in public subnet will be public
   - Manage internet gateway, route tables, and so on
2. Use VPC Peering
   - Create and manage different peering relationships (could be 100s)
3. Best way to expose a service VPC to your VPC is by using AWS PrivateLink
   - Simplifies network management and security of VPC
   - Requires a network load balancer on the service VPC and an Elastic Network Interface on the your VPCs
   - Like VPC peering but done at scale
   - Routes traffic between VPCs and services over the AWS network

AWS PrivateLink



https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/aws-privatelink.html

Direct Connect

What is Direct Connect?

- Cloud service solution to establish a dedicated private network connection between on-premise data center to AWS
- Reduces network costs
- Increase bandwidth throughput
- Provides a more consistent network experience than internet-based connections

AWS Direct Connect

https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/aws-direct-connect.html

2 Types of Direct Connect Connections

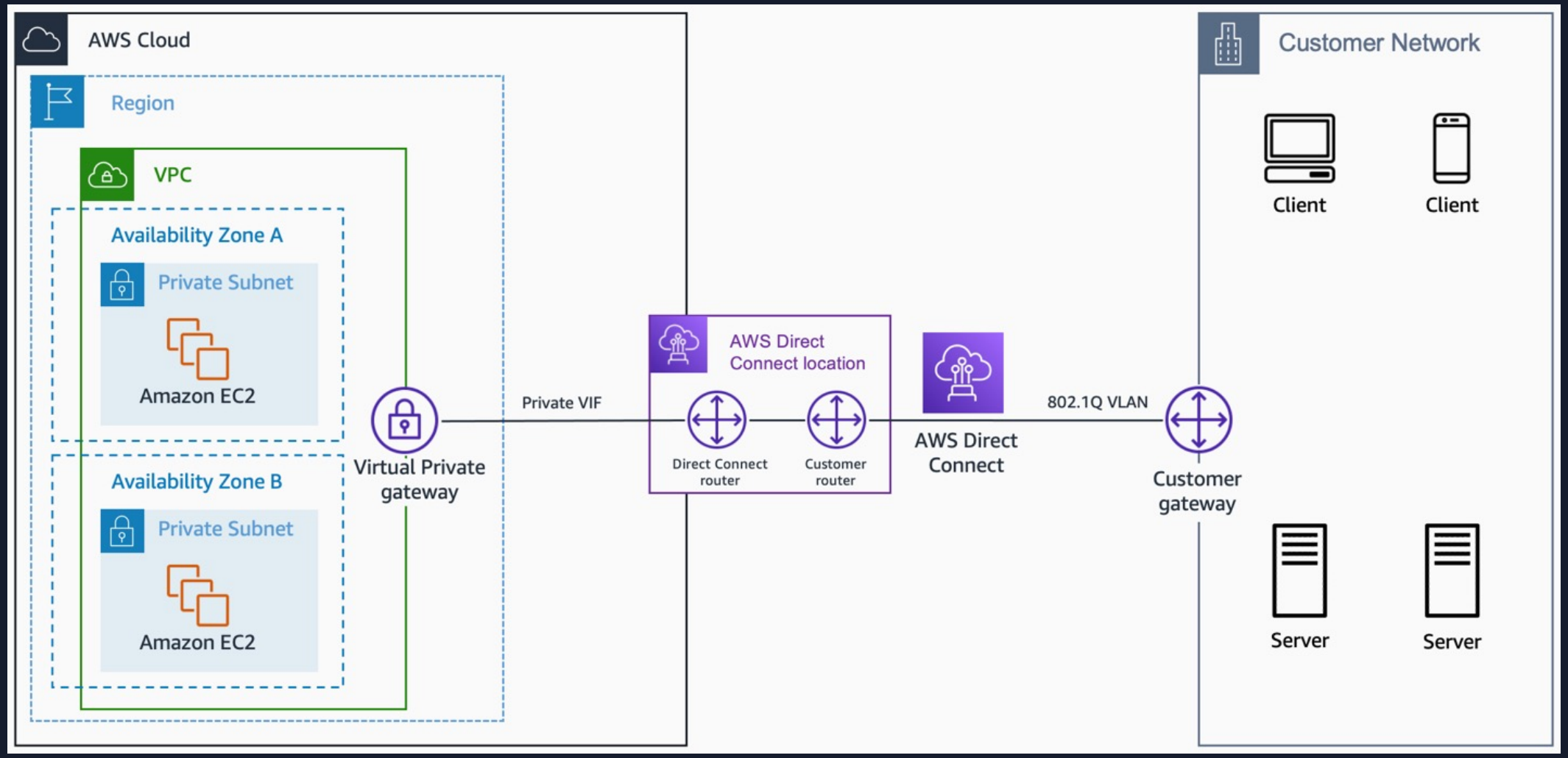
OPTION 1 Dedicated Connection

- Physical ethernet connection associated with a single customer
- Customers request a dedicated connection through the AWS Direct Connect console, CLI, or API

OPTION 2 Hosted Connection

- Physical ethernet connection that an AWS Direct Connect Partner provisions on behalf of a customer
- Customers request a hosted connection by contacting a partner in the AWS Direct Connect Partner program, who provisions the connection (e.g., Verizon, AT&T, …

AWS Direct Connect



Comparing VPNs to Direct Connect

VPN vs. Direct Connect

VPN

- Allows private communication
- Traverses the public internet to get the data delivered
- Secure but slow

VS

Direct Connect

- Fast
- Secure
- Reliable
- Able to handle massive throughput

Checkpoint

How would you scale an individual NAT gateway to support more bandwidth?

1. Stop the NAT gateway and resize it to a larger instance size.
2. You don't. AWS handles this for you.
3. Create a second NAT gateway and mirror the traffic.
4. Enable the Auto-Scaling option during creation.

Fill in the blank: Security groups are ____.

1. Stateish
2. Confusing
3. Stateful
4. Stateless

Fill in the blank: Network Access Control Lists (NACLs) are ____.

1. Stateish
2. Even more confusing than security groups
3. Stateless
4. Stateful

Which of the following is a reason that VPC peering could fail?

1. The VPCs have overlapping IP addresses.
2. The VPCs are in different accounts
3. The VPCs are in different regions.
4. The VPCs use different numbers of AZs.

By default, what range of IP addresses and ports do security groups leave open for inbound traffic?

1. None. They are closed by default.
2. 0.0.0.0/0, port 22
3. 127.0.0.1, port 3389
4. 0.0.0.0/0, port 80

Which of the following items are NOT a part of a VPC?

1. Internet Gateway
2. Service Gateway
3. Route Table
4. Subnet

What is the purpose of a NAT gateway?

1. To act as a firewall for your resources
2. To forward traffic to other EC2 instances
3. To block malicious traffic from leaving your network
4. To forward traffic from private subnets to services outside your VPC

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When would you want to use Direct Connect?

1. When you need a high-speed private connection from your on-premises environment to AWS
2. When you need to talk directly to AWS and don't want to use the normal support channels for a request
3. When your EC2 instance needs to enable enhanced networking
4. When you need your database to be able to move from one region to another

Your security team will only approve S3 usage if your EC2 instances don't transmit data over the public internet. What service can you use to comply with this requirement?

1. NACL
2. VPC Endpoint
3. Route Table
4. NAT Gateway

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